A Few Notes Harmonic Oscillators and Fields

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Last update: December 15, 2021

1 Introduction

This story begins somewhere around 1658 when Robert Hooke [2] began experimenting with springs and masses. By 1660 Hooke had made two significant steps, namely the use of a balance controlled by a spiral spring and an improved escapement called the anchor escapement. In 1660 he discovered an instance of what we now call Hooke's Law while working on designs for the balance springs of clocks. However he only announced the general law of elasticity in his lecture Of Spring given in 1678 [1].

Interestingly, it was the addition of the balance spring to the balance wheel around 1658 by Robert Hooke (and Christiaan Huygens [6]) that greatly increased the accuracy of portable timepieces, transforming early pocket watches from expensive novelties to useful timekeepers.

These notes are organized (I hope) as follows: Section 2 looks at Hooke's Law in some detail. Section 3.1 considers the simple pendulum and Section 3.2 considers the RLC circuits as examples of other simple harmonic oscillators. Section 4 looks at the Quantum Harmonic Oscillator. Section 5 outlines how quantum fields fit into all of this. Finally, 6 offers a few observations and conclusions.

2 Hooke's Law

Hooke's law [8] is a law of physics that states that the force (I'll just call it F) needed to extend or compress a spring by some distance (x) scales linearly with respect to that distance, that is, $F_s = -kx$, where k is a constant factor characteristic of the spring (i.e., its stiffness), and x is small compared to the total possible deformation of the spring. The law is named after 17th-century British physicist Robert Hooke [2]. He first stated the law in 1676 as a Latin anagram, and he published the solution of his anagram in 1678 as: ut tensio, sic vis ("as the extension, so the force" or "the extension is proportional to the force") [1].

2.1 The Mathematics of Hooke's Law

Hooke's Law is most frequently stated as $F \propto -x$, that is, the force F is proportional to the displacement x. However, more frequently this relationship is changed into an equality by the use of a spring constant k so that we have[1]

$$F = -kx \tag{1}$$

We also know that Newton's Second Law of Motion [9] tells us that

$$F = ma \tag{2}$$

[1] It seems to be convention to use x and $x(t)$ and a and $a(t)$ interchangeably.

Now, if we set Equation 1 equal to Equation 2 we see that

$$
\begin{aligned}
ma &= -kx && \text{\# Newton's force equals Hooke's force} \\
&\Rightarrow \quad a = -\frac{k}{m}x && \text{\# solve for } a\text{, noting that } a \text{ and } x \text{ are functions of time} \\
&\Rightarrow \quad \frac{d^2x}{dt^2} = -\frac{k}{m}x && \text{\#} \; a = \frac{d}{dt}(v) = \frac{d}{dt}\left(\frac{dx}{dt}\right) = \frac{d^2x}{dt^2}
\end{aligned}
$$

This situation is shown in Figure 1.

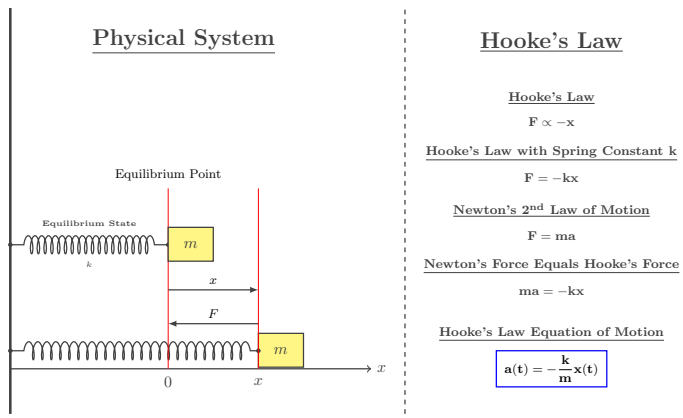

Figure 1: Spring and Mass System with Spring Constant k and Mass m

So now we know that

$$\frac{d^2x}{dt^2} = -\frac{k}{m}x \tag{3}$$

The next question is how do we solve the second order ordinary differential equation (Equation 3) for $x(t)$? Well, if we imagine the displacement ($x(t)$) plotted on the y axis against t (plotted on the x axis) we see the familiar pattern shown in Figure 2.

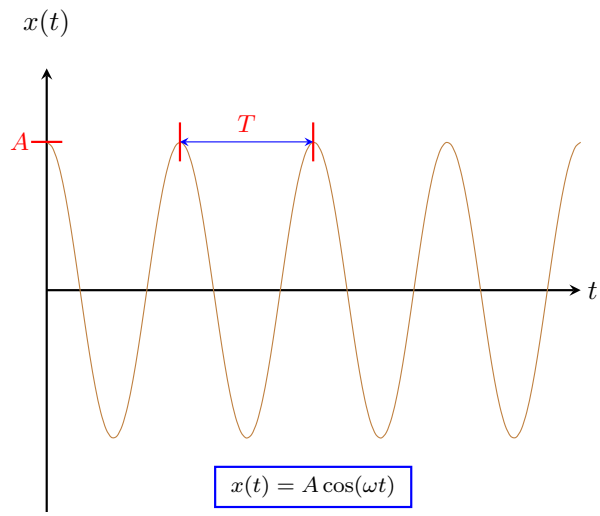

Figure 2: Simple Harmonic Motion Displacement Function

Here we can think of the spring being stretched to its maximum displacement (A) and then let go at time $t = 0$. The mass (m) will continue oscillating since in the Hooke's Law setup the only force acting on the mass is the force due to the spring, that is, $-kx$. This force is frequently referred to as the *restoring* force and is shown in Figure 1.

We can check to see if our guess, $x(t) = A\cos(\omega t)$, is really a solution to Equation 3 as follows:

$$
\begin{aligned}
\frac{d^2x}{dt^2} &= -\frac{k}{m}x && \text{\# Equation 3} \\
&\Rightarrow \frac{d^2}{dt^2}\left[A\cos(\omega t)\right] = -\frac{k}{m}x && \text{\# guess that } x(t) = A\cos(\omega t) \\
&\Rightarrow \frac{d}{dt}\left[-\omega A\sin(\omega t)\right] = -\frac{k}{m}x && \text{\# } \frac{d}{dt}\cos(u) = -\sin(u)\frac{du}{dt} \text{ with } u = \omega t \\
&\Rightarrow -\omega^2 A\cos(\omega t) = -\frac{k}{m}x && \text{\# } \frac{d}{dt}\sin(u) = \cos(u)\frac{du}{dt} \text{ with } u = \omega t \\
&\Rightarrow \omega^2 A\cos(\omega t) = \frac{k}{m}x && \text{\# cancel minus} \\
&\Rightarrow \omega^2 x = \frac{k}{m}x && \text{\# } x(t) = A\cos(\omega t) \\
&\Rightarrow \omega^2 = \frac{k}{m} && \text{\# cancel } x \\
&\Rightarrow \omega = \sqrt{\frac{k}{m}} && \text{\# take the square root of both sides}
\end{aligned}
$$

So $x(t) = A\cos(\omega t)$ is indeed a solution to Equation 3 in the case that

$$\omega = \sqrt{\frac{k}{m}} \tag{4}$$

Here A is the maximum displacement (or amplitude) and ω is the angular frequency (in radians per second). We also know that the definition of angular frequency is

$$\omega = \frac{2\pi}{T} = 2\pi f \tag{5}$$

where T is the period (time for one cycle in seconds) and f is the frequency in cycles/second [5]. Note that $f = \frac{1}{T}$ so f has units of seconds^{-1}.

Give this information we can write the displacement function $x(t)$ as in terms of the frequency f as follows:

$$x(t) = A\cos(2\pi f t)$$

We can also put Equations 4 and 5 together to find

$$f = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$$

and

$$T = 2\pi\sqrt{\frac{m}{k}}$$

2.2 Total Energy of a Simple Harmonic Oscillator

We know that the total energy of an oscillator such as the spring and mass system we looked at above is

$$E_{\text{total}} = \text{KE} + \text{PE}$$

All good, but what is the kinetic energy (KE) and potential energy (PE) of such as system?

Well, we know that [3]

$$E_{\text{total}} = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$$

so that

$$
\begin{aligned}
E_{\text{total}} &= \frac{1}{2}mv^2 + \frac{1}{2}kx^2 && \# \; E_{\text{total}} = \text{KE} + \text{PE} \\
&= \frac{1}{2}m\left(\frac{dx}{dt}\right)^2 + \frac{1}{2}kx^2 && \# \; v = \frac{dx}{dt} \\
&= \frac{1}{2}m\left(\frac{d}{dt}\left[A\cos(\omega t)\right]\right)^2 + \frac{1}{2}kx^2 && \# \; x(t) = A\cos(\omega t) \\
&= \frac{1}{2}m\left(-\omega A\sin(\omega t)\right)^2 + \frac{1}{2}kx^2 && \# \; \frac{d}{dt}\cos(u) = -\sin(u)\frac{du}{dt} \text{ with } u = \omega t \\
&= \frac{1}{2}m\omega^2 A^2\sin^2(\omega t) + \frac{1}{2}kx^2 && \# \; \left(-\omega A\sin(\omega t)\right)^2 = \omega^2 A^2\sin^2(\omega t) \\
&= \frac{1}{2}m\omega^2 A^2\sin^2(\omega t) + \frac{1}{2}k\left(A\cos(\omega t)\right)^2 && \# \; x(t) = A\cos(\omega t) \\
&= \frac{1}{2}m\omega^2 A^2\sin^2(\omega t) + \frac{1}{2}kA^2\cos^2(\omega t) && \# \; \left(A\cos(\omega t)\right)^2 = A^2\cos^2(\omega t) \\
&= \frac{1}{2}m\omega^2 A^2\sin^2(\omega t) + \frac{1}{2}m\omega^2 A^2\cos^2(\omega t) && \# \; \omega = \sqrt{\frac{k}{m}} \Rightarrow k = m\omega^2 \text{ (Equation 4)} \\
&= \frac{1}{2}m\omega^2 A^2\left(\sin^2(\omega t) + \cos^2(\omega t)\right) && \# \text{ factor out } \frac{1}{2}m\omega^2 A^2 \\
&= \frac{1}{2}m\omega^2 A^2 && \# \; \sin^2(\omega t) + \cos^2(\omega t) = 1 \\
&= \frac{1}{2}m(2\pi f)^2 A^2 && \# \; \omega = 2\pi f \\
&= \frac{1}{2}4\pi^2 f^2 A^2 m && \# \text{ expand } (2\pi f)^2 \text{ and rearrange} \\
&= 2\pi^2 f^2 A^2 m && \# \; E_{\text{total}} = 2\pi^2 f^2 A^2 m \\
&\Rightarrow E_{\text{total}} \propto f^2 A^2 && \# \; E_{\text{total}} \text{ is proportional to } f^2 A^2
\end{aligned}
$$

So now we know that the oscillator has $E_{\text{total}} = 2\pi^2 f^2 A^2 m$, or said another way $E_{\text{total}} \propto f^2 A^2$. This result will become useful when we consider the quantum harmonic oscillator.

3 Other Simple Harmonic Oscillators

3.1 The Simple Pendulum

A pendulum is a weight (that is, a mass m in a gravitational field) suspended from a pivot so that it can swing freely [10]. When a pendulum is displaced sideways from its resting, equilibrium position, it is subject to a restoring force due to gravity that will accelerate it back toward the equilibrium position. When released, the restoring force acting on the pendulum's mass causes it to oscillate about the equilibrium position, swinging back and forth. The time for one complete cycle, a left swing and a right swing, is called the period (T). The period depends on the length of the pendulum and also to a slight degree on the amplitude (A), the width of the pendulum's swing. The free body diagram [7] for the simple pendulum is shown in Figure 3.

From the first scientific investigations of the pendulum around 1602 by Galileo Galilei, the regular motion of pendulums was used for timekeeping, and was the world's most accurate timekeeping technology until the 1930 [4]. The pendulum clock invented by Christiaan Huygens in 1658 became the world's standard

timekeeper [6], used in homes and offices for 270 years, and achieved accuracy of about one second per year before it was superseded as a time standard by the quartz clock in the 1930s. Pendulums are also used in scientific instruments such as accelerometers and seismometers. Historically they were used as gravimeters to measure the acceleration of gravity in geo-physical surveys, and even as a standard of length. The word "pendulum" is new Latin, from the Latin *pendulus* meaning 'hanging' [11].

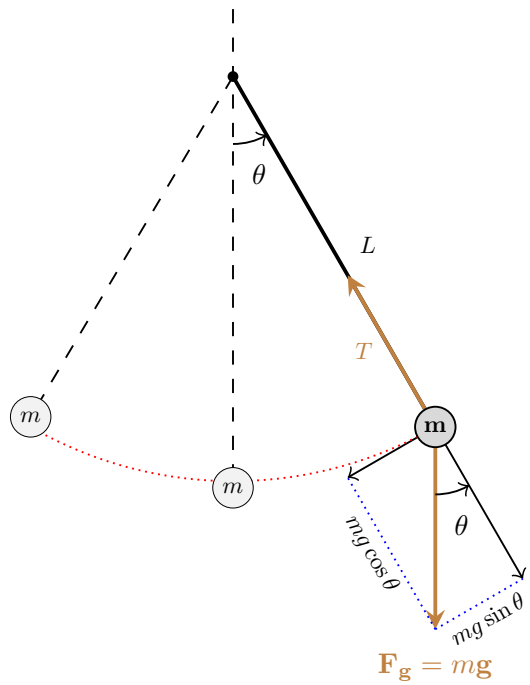

Figure 3: Free Body Diagram for a Simple Pendulum

3.2 The RLC Circuit

4 The Quantum Harmonic Oscillator

5 Quantum Fields

6 Conclusions

Acknowledgements

Thanks to Chris Lonvick (lonvick@gmail.com) for his comments and help on/with Figure 3.

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Figures

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